What the ALMA Release of 6 November 2026 Can Test: Pre-Data Predictions for U4_28473 and U4_27928

Working note, v1.0 (2026-11-01). AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted. The theory is not closed.

Abstract

Observing unit uid://A001/X3788/Xbfd2 of ALMA programme 2024.1.00100.L (CONDOR) is listed to become public at 02:42:31 UTC on 2026-11-06. We state this programme's predictions for it before any of its data can be read.

The unit's science target is U4_28473, at $z = 2.52$, not U4_27928, the galaxy this programme sealed for this date. U4_27928 lies 7.36″ off the phase centre, so its reading is serendipitous. At its empirical gas mass the frozen aperture gives S/N 0.16, so we expect that test to score nothing.

U4_28473 has no earlier seal, so we add one here. Its published CO(3–2) flux gives $\log M_{\rm gas} = 11.33$ ($\alpha_{\rm CO} = 4.36$) or 10.69 ($\alpha_{\rm CO} = 1$). At 3 kpc the baryons alone, under Newtonian gravity, reach 431–846 $\mathrm{km\,s^{-1}}$ across the input branches. The galaxy therefore tests its mass budget, not $a_0(z)$: where the CO reaches, g/a_0 is large and the readings differ by far less than the stellar-mass and $\alpha_{\rm CO}$ ranges. One disc excludes no model.

1 The framework, as it stands

The programme studies a galaxy law $g_N = g^2/(g + a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, on two footings never pooled ($a_0(0) = 9.3603 \times 10^{-11}$ and $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$), and a gravitating $w \simeq 0$ field component called cold dark energy, which is neither matter nor a particle. The working halo rule (N) sets a core $S_{\rm cold} = f_c\,\eta(M_t)\,\Sigma_{\rm coll}(M_t)$ with no a_0 in it.

The record's current position is unfavourable and is stated here so the predictions are read in context. On measured gas no tested rule fits galaxies, clusters and lensing together with what has been derived. The law remains the best-performing galaxy description, with misses of its own (Milky Way K_z, IC 1613), and has no cluster fit. Rule N fails dwarfs and has a hand-chosen core. Nothing in this note changes that, and nothing here says the data favour the framework.

2 What is released

From the ALMA archive (ObsCore query, 2026-11-01): band 3, beam 0.066″, 22 680 s on source. Windows 94.79–96.67, 97.60–98.62, 106.75–108.73 and 108.74–110.61 GHz, with line sensitivity 0.42–0.45 $\mathrm{mJy\,beam^{-1}}$ per $10\,\mathrm{km\,s^{-1}}$. The second window is centred on CO(3–2) at $z = 2.52$ (98.11 GHz). The dates and rights are the archive's and can change.

Correction to our own record. Our survey (CFG570) matched galaxies to archive rows by field of view. It listed U4_27928 as having CO(2–1) in this unit, and later files treated the unit as if aimed at it. KMOS$^{\rm 3D}$'s U4_28473 lies 0.002″ from the phase centre; U4_27928 lies 7.361″ away, at primary-beam response 0.96 for a 61″ beam. Its CO(2–1), at 95.29 GHz, does fall in the first window. The sealed test stands as written, as an off-axis reading. Whether the archive's imaged product extends to this offset at this resolution is unverified until release.

3 U4_27928: sealed requirements at the Hα speed

CFG571 (SHA-256 be27457d…) sealed the gas each model requires as a function of the measured V_c at 8.27 kpc. Addenda added $n = 1$, $n = 2$ and an AQUAL disc as labelled readings. None of these is altered here. Our Hα fit gives $v_{\rm rot} = 226.1$, $\sigma_0 = 77.8\,\mathrm{km\,s^{-1}}$, with a recorded median 28 % slow bias at the outermost radius. That gives V_c from 226 to $373\,\mathrm{km\,s^{-1}}$ depending on two branches never pooled, pressure support and bias (Table 1). The empirical scaling line used in the record gives $\log M_{\rm gas} = 10.51$.

Forecast. In the frozen $2R_e$ aperture over $\pm 500\,\mathrm{km\,s^{-1}}$, corrected for the primary beam, an extended disc reads S/N 0.16 at $\log M_{\rm gas} = 10.51$ and 0.49 at 11.0. The gate needs 5. The 3σ limit is $\log M_{\rm gas} = 11.79$ for $\alpha_{\rm CO} = 4.36$, above every framework requirement (at most 11.33), and 11.15 for $\alpha_{\rm CO} = 1$, below the requirement only in the fastest case. **Prediction A: G1 fails and no model is scored on this disc.** A much more compact or clumpy disc would change this.

Table 1: U4_27928: required total $\log M_{\rm gas}$ at the sealed $R = 8.27\,$kpc, $R_g = R_d$, canonical / alt, at the Hα speed. FLOOR: the stars alone already exceed the requirement. Route N: lg M_{200} 12.5, added after the criteria, information only.

case	V_c	n=1 F-DESI	n=1 F-flat	$\nu_{\rm mono}$ F-DESI	n=2 F-DESI	n=1 R-H	route N
a P0	226	9.72 / 9.22	9.62 / 8.66	9.89 / 9.57	10.17 / 10.07	FLOOR / FLOOR	10.35
b P0 /0.72	314	10.91 / 10.88	10.90 / 10.87	10.94 / 10.92	10.97 / 10.96	10.72 / 10.66	11.01
c P-B	303	10.84 / 10.81	10.84 / 10.80	10.88 / 10.85	10.92 / 10.90	10.63 / 10.55	10.96
d P-B /0.72	373	11.17 / 11.15	11.17 / 11.15	11.19 / 11.18	11.21 / 11.20	11.06 / 11.02	11.23

4 U4_28473: new predictions

Inputs, never pooled. $z = 2.52455$ (KMOS^{3D}). CO(3–2) $0.79 \pm 0.08\,$Jy km s^{-1}, gas R_e 1.56 kpc, stellar R_e 1.37 kpc at 4.4 μm (Tadaki et al. 2023, arXiv:2309.02703), with their $R_{31} = 1.8$. $\log M_\star$ 11.31 (Tadaki et al. 2017) and 11.35 (KMOS^{3D}), each also $-0.2\,$dex. Stellar R_e 3.16 kpc as a second shape, from KMOS^{3D}'s rest-optical 0.383″. Distances: flat, $h = 0.6856$, $\Omega_m = 0.2976$. These public numbers were read before the predictions were made, so this is a forward prediction of unreleased kinematics, not a blind test of the mass inputs.

Readings. $n = 1$ (canonical, primary), $\nu_{\rm mono}$, $n = 2$, the v3 AQUAL disc, route N at lg M_{200} 12–13, and Newton. For $a_0(z)/a_0(0)$: flat 1, F-DESI 0.819, and the ends of our own BAO band, 0.325 and 1.212 (quoted at $z = 2.33$, extrapolated). Two rivals as sealed: $a_0 \propto H(z)$ (R-H, 3.805) and the feedback RAR-equivalent (L-fb, 4.88, held at its table's end).

Table 2: U4_28473: predicted V_c in km s^{-1} at $R = 2/3/4\,$kpc from the published CO flux, canonical footing. S1: stellar R_e 1.37 kpc; S2: 3.16 kpc. Gas R_e 1.56 kpc.

lg $M_\star$, shape, $\alpha_{\rm CO}$	Newton	n=1 F-flat	n=1 F-DESI	n=1 R-H	route N 12.5
11.31, S1, 4.36	891/827/733	894/832/741	893/831/740	903/845/761	891/827/734
11.31, S1, 1	707/648/571	711/655/581	710/654/579	721/672/604	707/649/572
11.31, S2, 4.36	720/716/675	724/722/683	723/721/682	734/737/704	720/716/676
11.31, S2, 1	473/499/494	479/508/505	478/506/503	494/528/531	473/500/495
11.11, S1, 4.36	801/746/664	805/752/672	804/751/671	814/767/694	801/747/665
11.11, S1, 1	589/542/479	594/550/490	593/549/488	607/569/517	590/543/480
11.11, S2, 4.36	684/670/624	688/676/632	687/675/631	699/693/655	684/671/624
11.11, S2, 1	416/431/421	423/441/433	422/439/431	439/463/462	417/432/422

Table 3: U4_28473: $\alpha_{\rm CO,req}$ at $R = 3\,$kpc for a measured V_c, $n = 1$, canonical. F = FLOOR, stars alone exceed the requirement. The primary statistic of the scoring plan.

V_c	lg $M_\star$ 11.31, S2			lg $M_\star$ 11.11, S2			lg $M_\star$ 11.11, S1		
	flat	F-DESI	R-H	flat	F-DESI	R-H	flat	F-DESI	R-H
350	F	F	F	0.08	0.10	F	F	F	F
400	F	F	F	0.56	0.58	0.32	F	F	F
450	0.30	0.32	0.04	1.10	1.12	0.85	F	F	F
500	0.90	0.92	0.64	1.71	1.73	1.44	0.33	0.35	0.07
550	1.57	1.59	1.30	2.38	2.40	2.11	1.00	1.02	0.73
600	2.31	2.33	2.03	3.11	3.13	2.84	1.73	1.75	1.46

The AQUAL disc solution at lg $M_\star$ 11.31, S1 differs from the algebraic $n = 1$ in-plane value by a factor 0.990–1.007 at 1–8 kpc on the flat, F-DESI and R-H cells, so it adds nothing separable. Route N's core gives $V_{\rm core}(3\,{\rm kpc})$ about 25 km s^{-1} at lg M_{200} 12.5. Rule N has no derived redshift dependence; holding $\Sigma_{\rm coll}$ at its $z = 0$ value is an extrapolation, supported only by a halo acceleration ratio of 0.97–1.06 at $z = 2.33$ at fixed M_{200}.

Detection forecast, smooth disc. Beam 0.54 kpc. Peak S/N 6.4 per beam per 100 km s^{-1}; 3σ per beam out to 0.70 kpc; 3σ per azimuthal ring out to 2.21 kpc. Moment-0 aperture S/N over $\pm 500\,$km s^{-1} peaks at only 1.86, while a rotation-matched three-dimensional statistic reads 9.6. Clumps would raise the peak values.

Table 4: U4_28473: $V_c(3\,\mathrm{kpc})$ in $\mathrm{km\,s^{-1}}$ for one input branch (lg $M_\star$ 11.11, S2, $\alpha_{\mathrm{CO}} = 1$), every reading.

reading	canonical	alt
Newton	431	
$n = 1$, flat	441	443
$n = 1$, F-DESI	439	441
$n = 1$, band 0.325	435	435
$n = 1$, band 1.212	443	445
$n = 1$, R-H	463	469
$n = 1$, L-fb	471	477
ν_{mono}, flat	438	440
ν_{mono}, F-DESI	437	438
$n = 2$, flat	436	437
$n = 2$, F-DESI	435	436
route N, lg M_{200} 12.0	432	
route N, lg M_{200} 12.5	432	
route N, lg M_{200} 13.0	432	

Statements fixed before the data

1. **Prediction B1 (every reading with $\nu \geq 1$, and route N).** $V_c(R)$ at or above the Newtonian value of the true baryons. At 3 kpc the lowest branch gives $431\,\mathrm{km\,s^{-1}}$ (lg $M_\star$ 11.11, S2, $\alpha_{\mathrm{CO}} = 1$). A lower V_c requires lower $M_\star$, a more extended stellar disc, $\alpha_{\mathrm{CO}} < 1$, or missing pressure support, and it is not a statement about a_0.
2. **Prediction B2.** If $V_c(3\,\mathrm{kpc})$ is near the $430\,\mathrm{km\,s^{-1}}$ of our low-S/N Hα fit, $\alpha_{\mathrm{CO,req}}$ is 0.88 for lg $M_\star$ 11.11, S2, 0.07 for 11.31, S2, and FLOOR for every S1 branch ($n = 1$, flat, canonical). $\alpha_{\mathrm{CO}} = 4.36$ with published masses is then out for every reading. The Hα fit is weak: flagged low-S/N, edge-limited, dispersion at the prior's bound.
3. **Prediction B3.** No pair among flat, F-DESI and the band will be separated. At the branch of Table 4, flat and F-DESI differ by $2\,\mathrm{km\,s^{-1}}$ at 3 kpc.
4. **Prediction B4.** R-H sits $22\,\mathrm{km\,s^{-1}}$ above flat at 3 kpc on the same branch, while the mass branches span hundreds. No separation from R-H is expected either, unless the masses are fixed independently.
5. **Prediction B5.** The law exceeds route N by $9\,\mathrm{km\,s^{-1}}$ at 3 kpc and by 22 at 8 kpc on this branch. The forecast does not reach 8 kpc, so this release does not separate them.

5 The other sealed test

GS4_24110 ([CI](2–1), programme 2025.1.01377.L, public 2026-11-28 by our earlier metadata) is unchanged (Table 5). A governing-plan analysis already found every $n = 1$ versus $n = 2$ pair at or below 0.51σ, and the expected outcome there is also an upper limit. Other CONDOR targets have later dates (UDS16442 2026-11-25, ID2993 2026-12-04, GMASS1084 2026-12-22) and carry no predictions here.

Table 5: GS4_24110 (public 2026-11-28), sealed before this note, unchanged: required $\log M_{\mathrm{gas}}$ at R_e, V_c $206\,\mathrm{km\,s^{-1}}$, $R_g = R_d$, canonical / alt.

lg $M_\star$	model	ν_{mono}	$n = 1$	v3 AQUAL disc
10.89	F-DESI	9.82 / 9.11	9.53 / FLOOR	9.96 / 9.66
10.89	F-flat	9.46 / FLOOR	FLOOR / FLOOR	9.77 / 9.10
10.89	R-H	FLOOR / FLOOR	FLOOR / FLOOR	FLOOR / FLOOR
10.76	F-DESI	10.43 / 10.33	10.37 / 10.27	10.46 / 10.39
10.76	F-flat	10.36 / 10.24	10.30 / 10.18	10.42 / 10.33
10.76	R-H	FLOOR / FLOOR	FLOOR / FLOOR	FLOOR / FLOOR

6 Scoring

Frozen in commit 57d3ac6b64, alone, before any script. CO(3–2) detected at $\geq 5\sigma$ against a selection-aware null, centroid within $0.5''$, z within 0.004. V_c from a thin-disc forward model, with inclination and pressure branches never pooled. The primary statistic is $\alpha_{\mathrm{CO,req}}$ at the measured V_c, with its interval from $V_c \pm 1\sigma$ and $M_\star \pm 0.2$ dex. A reading is *consistent* if that interval includes α_{CO} in [0.5, 6], *in tension* if it lies wholly outside, and FLOOR if stars alone exceed the requirement even at $M_\star - 0.2$ dex. Two a_0 readings are separated only if their requirements differ by more than the $M_\star$ interval on both. Pooling, branch choice after the data, exclusion statements and law-form statements from one disc are forbidden. U4_27928 stays under its own plan (CFG806).

7 Limits

- One disc is a demonstration. No model is excluded.
- The G1 wording cites a moment-0 procedure, whose forecast for a smooth disc is below the gate. Our written expectation that G1 passes is not supported as frozen. The record keeps both, and any change needs a dated addendum before the data.
- The scoring harness for U4_28473 is not written. Criteria require it before the cube is read; anything else is labelled a departure.
- Stellar masses and fluxes keep the authors' reductions, including their cosmology for distances. Rivals and CFG571's angular size are as sealed.
- The rotation at these radii may not trace circular speed; pressure branches stay separate. An unmodelled bulge or AGN is not included.
- The E, E-CCBH and BH-k_3 a_0 branches are not recomputed at this redshift.
- Two rows were added after the criteria as information: route N on U4_27928 and $\alpha_{\mathrm{CO,req}}$ at $430\,\mathrm{km\,s^{-1}}$. One script check failed first on a threshold not in the criteria; 82 cells matched exactly, and the correction is disclosed in the lane README.

8 Reproducibility

Lane `rebuild_2026_10_22/AC1_alma_condor_predictions`: criteria 57d3ac6b64, run 02ffa20dd8; all checks pass, and a mutation (ten times R_{31}) trips its check and exits 1. `ac1_results.json` SHA-256 bfdfed68.... Seals: CFG571 be27457d..., $n = 1$ 9802de0d..., $n = 2$ d64727b5..., AQUAL b5d65a5e.... An audit script ties every quoted number here to those files.

AI assistance: Anthropic's Claude (analysis, review, drafting). ALMA metadata: ALMA Science Archive. Sources: Tadaki et al. 2017, ApJ 834, 135; Tadaki et al. 2023, arXiv:2309.02703; Wisnioski et al. 2019 (KMOS^{3D}).

$\kappa = \frac{1}{2}$ fitted; footings never pooled; cold dark energy is required beyond galaxies and is not derived; not theory closed.